Chapter 2: Matrix Algebra

2.1 Matrix Operations

- If A is an $m \times n$ matrix — that is, a matrix with m rows and n columns — then the scalar entry in the ith row and jth column of A is denoted by a_{ij} and is called the (i,j)-entry of A.

$$
\begin{array}{cc}
 & \begin{array}{ccc} a_1 & a_j & a_n \end{array} \\
 & \left[\begin{array}{ccccc} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{array}\right] = A \\
\end{array}
$$

Row i is the row $a_{i1} \cdots a_{ij} \cdots a_{in}$.

Figure 1 Matrix notation

- Each column of A is a list of m real numbers, which identifies a vector in $\mathbb{R}^m$.
- The columns are denoted by $a_1, \ldots, a_n$ and the matrix A is written as $A = [a_1\ a_2 \ldots\ldots a_n]$

The **diagonal entries** in an $m \times n$ matrix $A = [a_{ij}]$ are $a_{11}, a_{22}, a_{33}, \ldots\ldots,$ and they form the **main diagonal** of A. A **diagonal matrix** is a square $n \times n$ matrix whose nondiagonal entries are 0. An example is the $n \times n$ identity matrix, I_n.

- An $m \times n$ matrix whose entries are all zero is a **zero matrix** and is written as 0.

Sums and Scalar Multiples

- We say that two matrices are **equal** if they have the same size (i.e. the same number of rows and the same number of columns) and if their corresponding columns are equal, which amounts to saying that their corresponding entries are equal.
- If A and B are $m \times n$ matrices, then the **sum** $A+B$ is the $m \times n$ matrix whose columns are the sums of the corresponding columns in A and B.
- Since vector addition of the columns is done entrywise, each entry in $A+B$ is the sum of the corresponding entries in A and B.
- The sum $A+B$ is defined only when A and B are the same size.